InP-Based Surface-Emitting Distributed Feedback Lasers Operating at 2004 nm

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***Abstract*— We demonstrate InP-based buried grating coupled surface-emitting distributed feedback (DFB) lasers designed to operate at a wavelength of 2004 nm. The laser structure consists of three InGaAsSb/InGaAs QWs, with a 5 μm wide double-channel ridge waveguide. A second-order semiconductor/semiconductor grating is used for in-plane feedback and vertical out-coupling. The single longitudinal mode emission wavelength of the fabricated laser can be adjusted from 2002.7 to 2006 nm without any mode hopping. High side-mode suppression ratio (SMSR) of at least 35 dB is achieved under all injection currents and temperature conditions. The edge output power reaches 19 mW, measured in continuous-wave (CW) mode at 10 °C. Simultaneously, the output power of surface emission reaches 8 mW.**

***Index Terms*— InP-based, quantum well lasers, surface emission, distributed feedback.**

I. Introduction

INFRARED semiconductor lasers emitting around 2 μm have attracted extensive attention for a wide range of applications such as free-space communication, molecular spectroscopy, trace gas sensing, and medical devices [1]–[4]. Especially for space-borne lidar systems for continuous global-scale measurements of CO_2 and other atmospheric greenhouse gases, the 2004 nm lasers become urgently demanded due to the large absorption coefficient and minimal sensitivity to variations in temperature and water vapor concentration at this wavelength [5]–[7]. Stable single longitudinal mode, high continuous wave (CW) power, good beam quality and potential for low-cost, high-volume production are all important factors required for practical applications. The materials operating in this wavelength can be grown on both gallium antimonide (GaSb) and indium phosphide (InP) substrates. In recent years, high-power laterally coupled distributed-feedback (LC-DFB) lasers have been reported with more than 40 mW output power based on strained InGaAsSb/AlGaAsSb quantum wells (QWs) on GaSb substrate [8], [9]. However, state-of-the-art single-mode InP-based diode lasers are limited to output powers of few mW due to difficulties of large lattice mismatch epitaxy and other loss mechanisms [10], [11]. Considering the advantages of low cost, high quality, and compatibility with mature device-fabrication processes developed for optical fiber communications [12], [13], the InP material system is increasingly preferred over the traditionally used GaSb system [14]. What's more, for several applications like VCSELs or silicon integrated devices, InP-based lasers are favorable since the InP material platform is monolithically compatible with many photonics integrated circuits. Last year, we reported the promising result that hundreds of milliwatts output power emission was realized by a Fabry-Perot laser diode [15], which illustrate the great potential of InP-based lasers in this wavelength range.

Surface-emitting lasers have smaller form factors which are more suitable for large-scale 2D integration. What is more, they can provide broad emission area which is beneficial to far-field improvement comparing with edge emission semiconductor lasers. For 1.3 or 1.55 μm wavelength, growing two distributed Bragg reflector (DBR) layer stacks on the bottom and top to form a vertical cavity and obtain surface emission is the traditional method [16]. Another solution is to use a second-order buried grating to fabricate complex-coupled DFB lasers. A second-order grating can provide in-plane feedback and vertical out coupling due to second-order diffraction and first-order surface emission mode [17]. Metal/semiconductor, air-metal/semiconductor or metal/metal surface second order gratings providing adequate coupling are added on the top of the semiconductor to realize surface emission [18]–[20]. Such devices have a simple fabrication process without regrowth technology. However, these lasers are well known to fundamentally favor lasing in an antisymmetric mode with a two-lobed beam pattern, since the antisymmetric mode has the least radiation losses and consequently lowest threshold gain [21].

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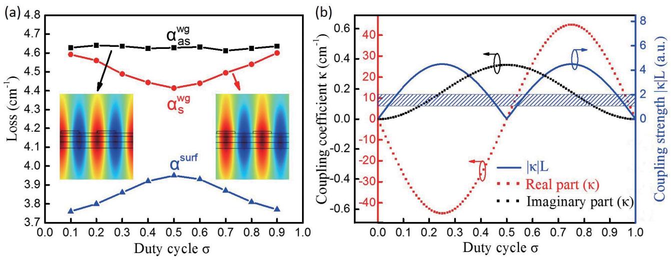

Fig. 1. (a) The calculated losses (α_s^{wg}, α_{as}^{wg} and α^{surf}) change with duty cycle σ. The inset shows modal profiles of symmetric and antisymmetric mode. (b) The coupling coefficient κ(grating depth = 40 nm) changes with duty cycle σ. Blue region indicates the optimized coupling strength range ($|\kappa|$L = 1∼2).

In principle, antisymmetric modes do not radiate in infinitely periodic structures, but become weakly radiative in finite-size devices, with low out-coupling efficiencies. Encouraged by the use of buried second-order grating technology, we apply semiconductor/semiconductor grating for surface emitting QWs laser for a low waveguide loss and high beam quality.

In this letter, we demonstrate the fabrication of 2004 nm wavelength InP-based DFB lasers. The emission wavelength can be finely controlled from 2002.7 nm to 2006 nm with a temperature-tuning rate of 0.11 nm/K. High side-mode suppression ratio (SMSR) of at least 35 dB is achieved under all injection currents and temperature conditions. Corresponding well with our simulation, a high output power of edge and surface emission with symmetric mode lasing was obtained. The maximum CW power of edge and substrate emission are as high as 19 mW and 8 mW at 10 °C, respectively, giving a much better performance than the earlier 2-μm wavelength InP-based ridge waveguide DFB lasers [11]. In addition, a single-lobed far-field radiation pattern with a low divergence of about 0.22° × 8.9° is achieved.

II. Device Design and Fabrication

According to the coupled-wave theory for gratings [17], the feedback in the DFB structure relies on periodic modulation of the refractive index. We solve for the complex eigenfrequencies of laser by modeling an infinite grating structure (depth = 40 nm) employing periodic boundary conditions and the grating period $\Lambda_g = \frac{m\lambda}{2\, n_{eff}}\left(\mathrm{m} = 2, \mathrm{n}_{eff} = 3.19\right)$ using finite-element method (FEM). Moreover, the material properties of active region and waveguide are specified in term of complex refractive index. Fig. 1(a) shows the waveguide losses of symmetric mode $\left(\alpha_s^{wg}\right)$, antisymmetric mode $\left(\alpha_{as}^{wg}\right)$ and surface emission loss $\left(\alpha^{\mathrm{surf}}\right)$ as a function of grating duty cycle (σ, defined by the ratio of grating teeth length and Λ_g). It is evident that the value of α_{as}^{wg} is always higher than α_s^{wg}, which means that the buried grating supports lasing of the symmetric mode. The maximum of mode loss difference appears at duty cycle $\sigma = 0.5$ with a value of 0.21 cm^{-1}, where the stability should be optimal. The inset images show modal profiles of symmetric and antisymmetric mode. In the asymmetric mode, lobes of opposite sign for the electric field appear at two interfaces where the index changes abruptly. So the field destructively interferes outside the grating with no surface radiating component. In contrast, the symmetric mode has single lobed electric field more concentrated below the grating teeth, i.e. the higher index part of the grating, and thus produces surface radiation because of constructive interference.

Then we calculated the complex coupling coefficient κ of the second order grating. The real and imaginary part of the coupling coefficient represent the second order and first order coupling coefficient, respectively. The values of κ can be calculated from the following equations derived from coupled mode theory [22].

$$k = -\frac{\Delta n_{eff}}{\lambda} \sin\left(2\pi d_c\right) + i\frac{4\Delta n_{eff}\,\Delta x d_g}{\lambda^2} \sin^2\left(2\pi d_c\right)$$

where Δn, $\mathrm{d_g}$, $\mathrm{d_c}$, and $\Delta \mathrm{n_{eff}}$ are the refractive index difference at the grating interface, the etch depth, the duty cycle, and the index modulation, respectively. These parameters are Δn = 0.1, $\mathrm{d_g}$ = 40 nm, and $\Delta \mathrm{n_{eff}}$ = 0.009. The values of real part (index-coupling) and imaginary part (loss-coupling) of κ as functions of grating duty cycle are shown in Fig. 1(b). For a surface emitting device, we would like to make the surface radiation component α^{surf} maximized. According to this, the grating duty cycle should be 50%. But the coupling strength is too weak at a duty cycle of 50%. Therefore we plotted the coupling strength of $|\kappa|$L for a 1 mm cavity length device as blue solid line. The blue region in Fig. 1(b) shows the optimum coupling strength of $|\kappa|$L = 1∼2 for high output power [17] and the corresponding range of the duty cycle is 42.5% to 46.2% or 53.9% to 57.5%. However, usually, the emission mode is unstable in this case, the F-P modes also start lasing due to mode competition. We expect to restrain the impact of longitudinal modes by increasing the coupling strength. Finally, we choose a duty cycle of about 33%, where the coupling strength of $|\kappa|$L is around 4, as a trade-off between surface radiation and stable single mode operation with high output power.

The laser structures were grown on 2-inch. S-doped InP substrates using an Aixtron close coupled showerhead (CCS) 3×2 Flip-Top metal organic chemical vapor deposition (MOCVD) system. A 300 nm Si-doped n$^+$-InP buffer was firstly deposited on the InP substrate. The active core was sandwiched between two InGaAsP layers, and the InGaAsP layers were surrounded by upper and bottom InP waveguide layers. The active core consists of three periods of $In_{0.63}Ga_{0.37}As_{0.99}Sb_{0.01}/In_{0.53}Ga_{0.47}As$ quantum wells and barriers, identical to that in our previously published paper [15]. The second order grating with a period $\Lambda_g = 628$ nm was fabricated on the upper InGaAsP confining layer. Fig. 2(a) shows the scanning electron microscopy (SEM) image of the second order grating. Here, the grating was fabricated by two-beam holographic lithography and dry etching to a depth of about 40 nm. According to our simulation, we chose the duty cycle near 33% which can be controlled by adjusting exposure and developing time, shown in Fig. 2(a). After the grating fabrication process, the 1200 nm p-InP upper cladding layer and the 200 nm Zn-doped p$^+$-InGaAs were regrown.

After regrowth, the wafer was patterned by photolithography and wet-etching into double-channel ridge waveguide lasers with an average core width of 5 μm as shown in Fig. 2(b).

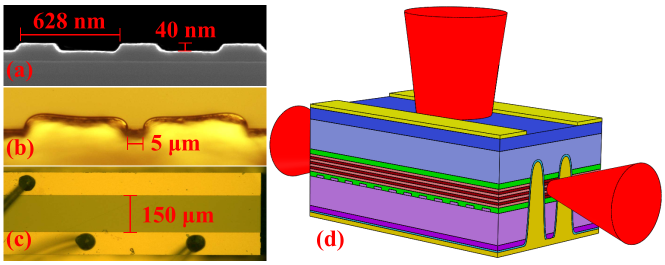

Fig. 2. (a) SEM image of the second order grating, (b) Optical microscopy images of cleaved cavity facet and (c) substrate emission window, (d) three-dimensional schematic presentation of the lasers.

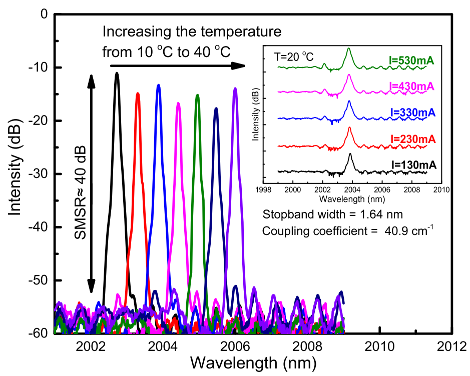

Fig. 3. CW emission spectra of a second-order DFB laser at different temperatures from 10 to 40 °C. The inset shows the lasing spectra under different injection currents at 20 °C.

Then a 450 nm thick SiO_2 layer was deposited by Plasma Enhanced Chemical Vapor Deposition (PECVD) for insulation around the ridges. After evaporation of the top non-alloyed Ti/Au contacts, an additional 5 μm thick gold layer was subsequently electroplated to further improve heat dissipation. After being thinned down to about 120 μm, a Ge/Au/Ni/Au metal contact layer was deposited on the substrate side of the wafer. 150 μm wide emission windows as shown in Fig. 2(c) were opened in the backside metal. The wafers were cleaved into 1 mm long bars. In addition, a single layer of 220 nm Al_2O_3 providing 6.7% reflectivity was deposited on the front facet to suppress mode hopping. The final lasers were mounted epilayer side-down on diamond heat-sink with indium solder. Fig. 2(d) presents the three-dimensional schematic image of the lasers.

III. Laser Characterization

The spectra were measured by a long wavelength optical spectrum analyzer with a resolution of 0.02 nm. Fig. 3 shows the CW surface emission spectra of a 1 mm long laser near 1.05 I_{th} drive current at different heat sink temperatures of 10–40 °C. A SMSR of up to 40 dB was achieved and no mode hopping was observed in the entire temperature range. The spectral peak was shifted from 2002.7 to 2006 nm, corresponding to a temperature tuning coefficient of 0.11 nm/K. The inset shows the lasing spectra at different currents between 130 mA and 530 mA in steps of 100 mA at 20 °C. The current tuning rate is lower than that reported in [11] where epi-up

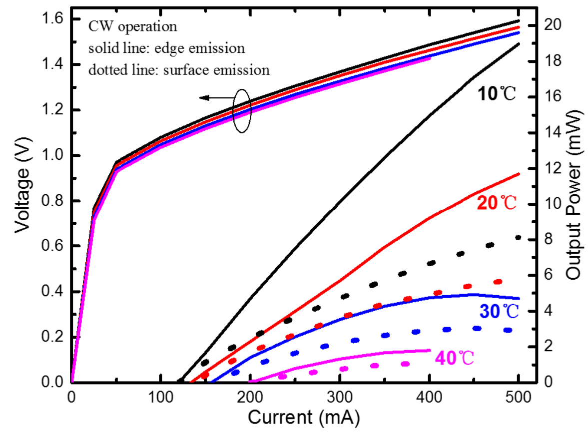

Fig. 4. L-I-V characterization at different heat sink temperatures of 10–40 °C for laser diode operated in CW mode.

mounting on copper is adopted, which indicates the lower thermal resistance obtained by epi-down mounting. Single longitudinal mode operation to high current was realized and the SMSR of spectrum was always higher than 35 dB for all injection currents. In fact, a certain amount of gain/loss coupling is always present in a second order DFB laser. This loss coupling originating from modulation of waveguide loss and surface radiation loss should prevent the device from oscillation in both stop-band modes. In addition, a stopband width of 1.64 nm was measured and we calculated the coupling coefficient $\kappa = \Delta \upsilon \cdot \pi \cdot n_{eff} = 40.9\ \text{cm}^{-1}$. For our 1 mm long devices, the coupling strength of $|\kappa|$L is 4.1 which corresponds well with our design.

The emitted optical power was measured with a calibrated thermopile detector placed right in front of the emission window or the coated laser facet for the surface and edge radiation measurement, respectively. Fig. 4 shows the CW light-current-voltage (L-I-V) curves of the laser at different heat sink temperatures. At 10 °C, the output power can reach 19 mW and 8 mW for edge and surface emission, respectively, which represents the best reported results so far for any InP-based surface emitting lasers at this wavelength. According to the equation $\frac{P_{\text{from}}}{P_{\text{back}}} = \frac{\left(1-R_f^2\right)\cdot R_b}{\left(1-R_b^2\right)\cdot R_f}$, we can deduce an output power of at least 5 mW from uncoated back cavity facet and the total power from three output facets can surely exceed 30 mW. Threshold power consumption is about 0.1 W, which is sufficient for the CO_2 gas detection by space-boren lidar systems [8]. In addition, we also notice that the laser's output power reduced to a few mW at 40 °C. This may be ascribed to the enhanced Auger recombination and more hot carriers escaping from the QWs at higher temperatures. The edge emission slope efficiency ∼0.06 W/A at a low injection current and the total wall-plug efficiency ∼ 4.3% can also be calculated from the above L-I-V curves.

The far-field profiles are presented in Fig. 5. The testing was done at a distance of 30 cm away from the laser. The full width at half maximum (FWHM) of the lateral far-field profiles is also indicated. Theoretical calculations show that the lowest threshold gain mode emitted through the surface in a second order semiconductor/semiconductor grating device is

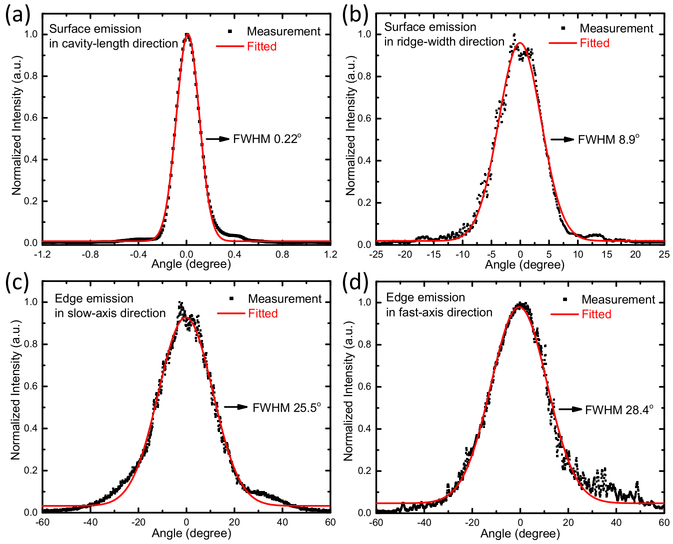

Fig. 5. Far-field profiles of surface emission in (a) cavity-length direction, (b) ridge-width direction, and for edge emission in (c) slow-axis direction, and (d) fast-axis direction. The dots are the measurement which is fitted with the Gauss functions shown as red lines.

the symmetric mode. Here we observed a single lobed pattern with a small beam divergence of about 0.22° FWHM in the direction along the cavity length, which is associated with the 1 mm wide longitudinal aperture. On the other hand, the beam is relatively divergent (8.9°) in the perpendicular direction due to a waveguide width that is of the same order as the wavelength, as shown in Fig. 5(b). Fundamental transverse mode emission from the edge with a near diffraction limited beam divergence about 25.5°× 28.4° has been confirmed, as shown in Fig. 5(c) and (d).

IV. Conclusion

In conclusion, we have demonstrated CW operation of surface emitting InP-based InGaAsSb/InGaAs QWs laser. The emission wavelength can be adjusted from 2002.7 to 2006 nm with a temperature-tuning rate of 0.11 nm/K. High side-mode suppression ratio (SMSR) is achieved with at least 35 dB under all injection currents and temperature conditions. At 10 °C, a total CW output power as high as 30 mW has been achieved. A single lobed surface-emitting far-field radiation pattern with a low divergence angle of about 0.22°× 8.9° is observed, which proves that the designed grating is reasonable and effective. These InP-based lasers will contribute a large step towards practical applications for CO_2 gas detection equipped in lidar system.

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